

Hamza ISLAH

PhD Candidate in Constraint Programming and Machine Learning

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Research Interests

Integration of machine learning and constraint programming: building systems that assist in modelling constraint problems, predicting the unsatisfiability of constraint problems, and using machine learning to improve CSP solving.

Education

2023 – present	PhD in Constraint Programming and Machine Learning <i>Mohammed VI Polytechnic University (UM6P), Morocco</i>
2020 – 2022	MSc in Numerical Methods and Applied Statistics <i>Faculty of Science Ain Chock, Hassan II University of Casablanca, Morocco</i>
2016 – 2022	BSc in Applied Mathematics <i>Faculty of Science and Technology, Errachidia, Moulay Ismail University, Morocco</i>

Publications

Peer-reviewed conference papers

1. **Islah, H.** and Benhaddou, D. (2024). Machine Learning Based Recommendation Queries for Constraint Acquisition. *2024 IEEE 36th International Conference on Tools with Artificial Intelligence (ICTAI)*. DOI: 10.1109/ICTAI62512.2024.00031.

Under review

1. **Islah, H.** and Mechqrane, Y. Recommendation Queries for Constraint Acquisition via Link Prediction. Under review.
2. **Islah, H.** and Mechqrane, Y. Graph Neural Networks for CSP Satisfiability Prediction. Under review.

Research and Professional Experience

Feb 2023 – May 2023	Data Science Intern <i>Shared System</i> • Conducted research in constraint programming and machine learning, focusing on optimisation problems and their applications.
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Teaching and Outreach

2025	Mentor, Ramadan AI Hackathon <i>Ministry of Digital Transition and Administration Reform</i>
2024	Instructor, Introduction to Programming and AI for Children <i>Ministry of Digital Transition and Administration Reform, Errachidia</i> Taught the fundamentals of constraint programming, including modelling and solving optimisation problems.

Technical Skills

- **Programming languages:** Python
- **Machine learning frameworks:** PyTorch, scikit-learn
- **Constraint programming tools:** OR-Tools
- **Data analysis and visualisation:** Pandas, NumPy, Matplotlib, Seaborn
- **Optimisation techniques:** Linear programming, integer programming, constraint programming

Languages

Arabic (native) · French (fluent) · English (fluent)